

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Problem Interpretation (2)	Analytical Reasoning (2.5)	Method Selection (2)	Solution Quality (2)	Presentation of Analysis (1.5)	TOTAL (10)
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Analytical Problem Solving

1. A cloud datacenter host has 64 CPU cores and supports CPU overcommitment at a ratio of 2.5:1. Each virtual machine (VM) requires 4 vCPUs. Calculate the maximum number of VMs that can be deployed

on this host. If actual CPU utilization reaches 85%, analyze whether further VM allocation is advisable.

2. A virtualized storage system provides 20 TB of physical storage with thin provisioning at a ratio of 3:1. Each VM is allocated 500 GB of virtual disk space. Calculate the maximum number of VMs that can be provisioned.
3. Evaluate the security implications of multi-tenancy in cloud datacenters. Analyze how virtualization can both enable and weaken isolation, and recommend mechanisms to strengthen tenant separation.
4. Analyze the role of Service-Oriented Architecture (SOA) in enabling interoperability across heterogeneous cloud platforms. Identify key challenges such as service latency, security, and version control, and propose solutions.
5. A cloud system implementing presence services faces scalability issues during peak usage. Analyze the architectural bottlenecks and recommend design improvements to ensure real-time updates and efficient resource utilization.

Question - 1 CPU Overcommitment calculation.

Given,

physical CPU cores = 64.

overcommitment ratio = 2.5:1.

Each VM requires = 4 vCPUs.

Total available vCPUs : $64 \times 2.5 = 160$ vCPUs.

Maximum number of VMs : $\frac{160}{4} = 40$ VMs.

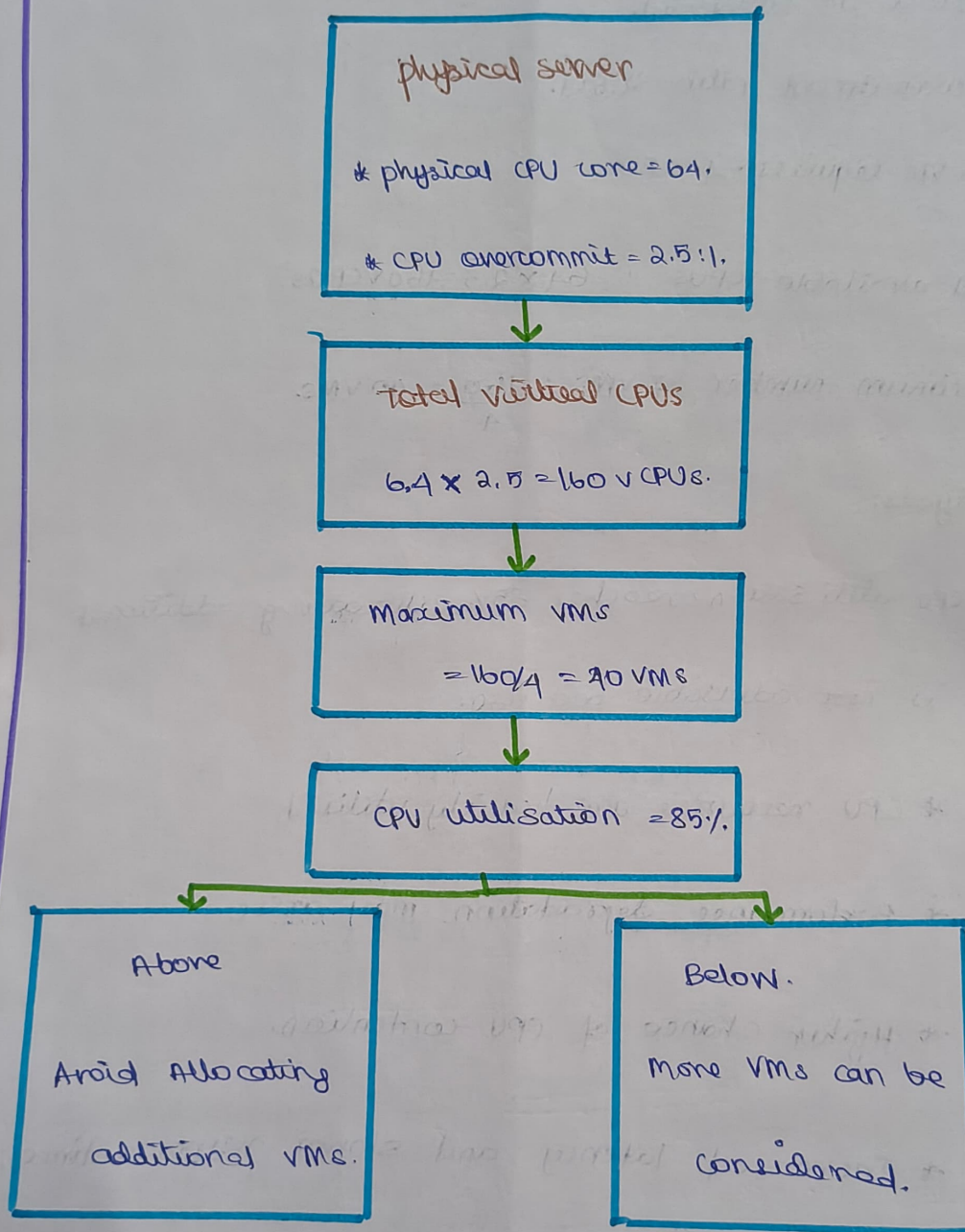
Analysis:

If CPU utilisation reaches 85%, allocating additional VMs is not advisable because:

- * CPU resources are heavily utilised.
- * Performance degradation may arise.
- * Higher chance of CPU contention.
- * Increased latency and slower response time.

Recommendation: maintain current allocation or migrate workloads to another host before creating additional VMs.

Cloud Datacenter Host.



Question 2:

Given,

physical storage

Thin provisioning

Each VM = 500 GB

virtual storage

Number of VMs

Maximum

Note:

Thin provis

full alloc

monitoring

Question 2: Thin Provisioning storage calculation.

Given,

physical storage = 20 TB.

Thin provisioning ratio = 3:1.

Each VM = 500 GB.

Virtual storage $\Rightarrow$

$$20 \times 3 = 60 \text{ TB}$$

$$60 \times 1000 = 60000 \text{ GB}$$

Number of VMs $\Rightarrow$

$$\frac{60000}{500} = 120.$$

Maximum number of VMs = 120.

Note:

Thin provisioning assumes that not every VM uses its full allocated storage simultaneously. Continuous monitoring is required to prevent storage exhaustion.

Question 3: Security Implications of Multi-tenancy.

How virtualisation enables isolation:

- * Each tenant runs into a independent virtual machine.
- * Separate tenant CPUs, memory, and storage are allocated.
- * Hypervisor isolates operating systems.
- * Prevents direct access between tenants.

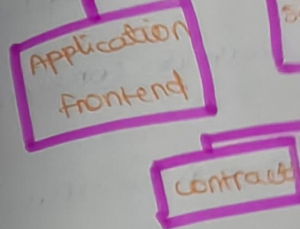
Security risks: * VM escape attacks.

- * side channel attacks.
- * Misconfigured virtual networks
- * Data leakage b/w tenants.
- * Shared resource vulnerability

Recommendations: * use secure Type-1 hypervisors.

- * Encrypt data at rest and in transit.
- * Enable network segmentation using VLANs.
- * Apply least privilege access control.
- * Regular patching and vulnerability scanning.
- * Use Multi-Factor Authentication (MFA).

Question 4:



Role of SOA: SOA allows

platforms to communicate

Benefits: * platform

* Reusable

* Loose

Challenges: * High

* Inter

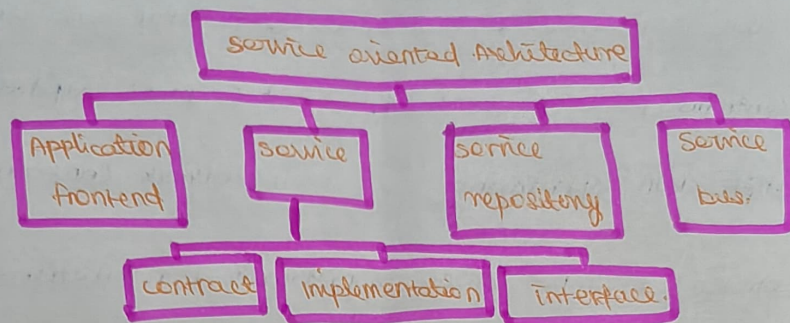
Solutions: * HTTP

* Monitoring

* API gateway

* Service registry

Question 4: Role of SOA in cloud Interoperability.



Role of SOA: SOA allows applications on different cloud platforms to communicate through standardised services.

Benefits: * Platform Independence. * Scaling.

* Reusable Services

* Easy Integration.

* Loose coupling.

Challenges: * High network latency * security vulnerabilities

* Interoperability issues * service version conflicts.

Solutions: * HTTPS encryption * Caching.

* Monitoring using cloud management tools.

* API gateway

* Load balancing

* OAuth Authentication

* service registry and discovery.

Question 5. Presence Service Scalability

Possible bottlenecks: * large number of simultaneous users.

* Centralised server overload. * frequent updates.

* High CRUD operations

* Network congestion.

Recommended Improvements: * Distributed architecture.

* Auto Scaling cloud Instances.

* Load Balancing.

* Event driven messaging.

* Database sharding.

* Monitoring with Prometheus and Grafana.

* Web sockets for real-time communication

Benefits: * Faster real time updates.

* Better scalability.

* Reduced server load.

* Lower response time.

* Improved utilisation.

Conclusion: A distributed, event driven architecture with

caching and auto-scaling ensures efficient handling of

large-scale presence services.